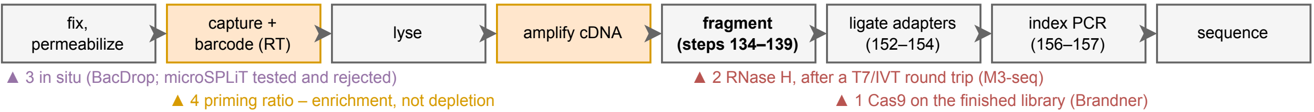
Ribosomal RNA depletion: four chemistries, four places in the workflow

About 85% of a yeast cell's RNA is ribosomal and an undepleted split-pool library spends roughly 94 of every 100 reads on it, so this is the largest single cost lever in the document. The four routes are not variants of one recipe: they act on different molecules at different points, and the one head-to-head comparison in the literature went against the most obvious of them. Colors and the 5′-to-3′ convention follow Figs. 2 to 4.

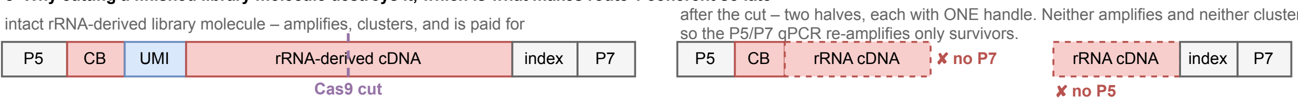
a Where each route acts. The library is only adapter-flanked, and therefore destroyable by a cut, after indexing.



b What each route does to the molecule, and what it measured

route	acts on	mechanism	measured outcome
1 Cas9 cleavage Brandner, mapSPLiT lineage: DASH, Prezza	finished library (post-index)	253 sgRNAs against 5S/16S/23S, weighted by the rRNA coverage their own runs produced, cut the rRNA cDNA. The mass stays in the tube; the cut molecules simply lose one flanking adapter and can no longer amplify or cluster. Depletion here is a selection.	E. coli 4.6% → 60.2% mRNA; median transcripts per cell 54 → 115. P. putida 2.2% → 10.4% with the SAME protocol, which is the transfer risk in one number.
2 RNase H digestion Wang, M3-seq	library, converted back to RNA	A T7 promoter is added by PCR and the library transcribed to ssRNA. DNA probes complementary to rRNA anneal, and thermostable RNase H cuts the RNA strand of each hybrid. DNase I removes the probes; a second RT returns the survivors to cDNA. The cut-the-adapters trick is unavailable here, because the molecule is RNA.	Chosen over Cas9 in the only head-to-head: "after testing two approaches … we chose an RNase H-based approach". Their Cas9 arm left 75–80% rRNA in the final library.
3 In situ, on the cell BacDrop; microSPLiT tested three	unamplified RNA, inside the cell	microSPLiT tried three and kept the weakest-sounding one: poly(A) polymerase, which tags mRNA rather than removing rRNA; a 5′-phosphate-dependent exonuclease; and probes plus RNase H. PAP won. BacDrop instead digests enzymatically before indexing, which M3-seq notes "risks digesting non-rRNA transcripts".	PAP gave "about 2.5-fold, or about 7% of total RNA" enrichment – the best in-situ result published, against 13-fold post hoc. M3-seq avoids the route because in-situ depletion "can decrease mRNA capture efficiency".
4 Prime differently Brettner, yeast (not depletion)	the RT primer mix	rRNA is not polyadenylated, so an oligo-dT primer does not reach it and the random hexamers do. Lowering the hexamer:dT ratio simply stops the assay reaching for rRNA. Nothing is destroyed, so this is enrichment, and it is free apart from a titration.	Not measured. Brettner et al. recommend it and do not run it, recovering 93.7% rRNA against 5.75% mRNA. It also shifts transcript-length bias, since hexamer priming scales with length (Sec. 2.4).

c Why cutting a finished library molecule destroys it, which is what makes route 1 coherent so late



Segments mRNA / cDNA rRNA enzyme or probe UMI handle / adapter dashed = destroyed or absent

The ordering that matters: route 1 acts AFTER fragmentation and AFTER indexing, not before, which an earlier version of this document had backwards. Brandner et al. state their input as the library "from after the final double-sided size selection"; in the microSPLiT protocol that is step 170, with fragmentation at 134–139 and index PCR at 156–157. Composed from the protocols as described in Secs. 3.1 and 3.5, not traced from any published figure. Every number drawn by hand here is sourced in the provenance table for this figure.